Flap/Elevator Interconnect Trim System – General Data

1. General

The flap/elevator interconnect trim system is a remote mechanical operation in which wing flap actuator travel is used to apply compensating nose down trim proportional to the amount of flap extension.

2. Description

The system comprises a ball screw jack, a cable system, trim tab screw jack, connecting rod and trim tab.

A. Ball Screw Jack

The flap/elevator interconnect ball screw jack, mounted in the cabin roof structure and connected to the flap operating mechanism, converts flap system linear motion into rotary motion. The ball screw jack comprises a ball screw assembly, a spindle and an end support. The ball screw assembly consists of a threaded shaft, which engages ball bearings enclosed in a tube integral with a ball cage. The ball cage is keyed into a spindle which has an integral cable drum, and the spindle rotates on a bearing mounted on the end support bracket. The end support bracket is bolted to the cabin roof structure and a fork end installed on the screw shaft connects to the flap operating mechanism. Linear movement of the screw shaft causes the ball cage and spindle to rotate and drive the cable drum.

B. Trim Tab Screw Jack

The flap/elevator interconnect trim tab screw jack mounted on the front spar of the right elevator, converts rotary motion into linear motion. The screw jack comprises a cable drum spindle which rotates in bearings in a stop cover, and a slide screw assembly. The cable drum is internally threaded and engages with external threads on the slide screw. A fork end is threaded and riveted into the slide screw.

C. Trim Tab

The flap/elevator interconnect trim tab is hinged to the trailing edge of the right elevator. A bracket on the lower surface of the tab provides for the connection of a connecting rod, from the screw jack slide screw fork end.

3. Operation

During flap operation, movement of the right inboard operating bellcrank (refer to 27-50-00, Wing Flaps System – General Data, Para 2. and Para 3.) causes the ball screw jack spindle and cable drum to rotate. The cables attached to the ball screw jack drum transmits this movement to turn the interconnect trim screw jack cable drum, which extends or retracts the screw jack slide screw and operates the interconnect trim tab through the connecting rod. For a flaps down selection, the interconnect trim tab moves up, to provide a nose down trim proportional to flap movement.

4. Removal/Installation

A. Install Flap/Elevator Interconnect System Rear Pulley Assembly

1) Install rear pulley bracket assembly, referring to, Figure 1 or Figure 2 for assembly details. Lubricate pulley bush (refer to 12-20-10, Lubrication – General Data). Ensure after tightening pivot bolt there is free movement of the bracket on the bushing.

2) Connect flap/elevator interconnect cables and rig trim system, refer to Para 5.A or Para 5.B.

5. Adjustment/Test

A. Rig Flap/Elevator Interconnect Trim System (Mod 6/1219)

Note Before attempting to rig the flap/elevator interconnect trim system, ensure that the wing flap system is correctly rigged. (Refer to 27-50-00, Wing Flaps System – General Data).

Refer to Figure 1 for details.

1) Ensure wing flaps are pumped fully up.

2) Set cables on interconnect ball screw jack cable drum as in Figure 1, "Detail A", with inboard cable terminal uppermost.

3) Set trim screw jack extension to between 0.030 and 0.070 inch and arrange cables on screw jack drum as shown in Figure 1, "Detail B".

4) Maintaining positions of ball screw jack and screw jack as in Para 5.A.2) and Para 5.A.3), adjust and tension trim cables at turnbuckles in rear fuselage. For cable tension values refer to 27-00-00, Flight Controls – General Data. Lock turnbuckles using locking clips.

5) Disconnect connecting rod between screw jack and tab attachment bracket, at tab attachment.

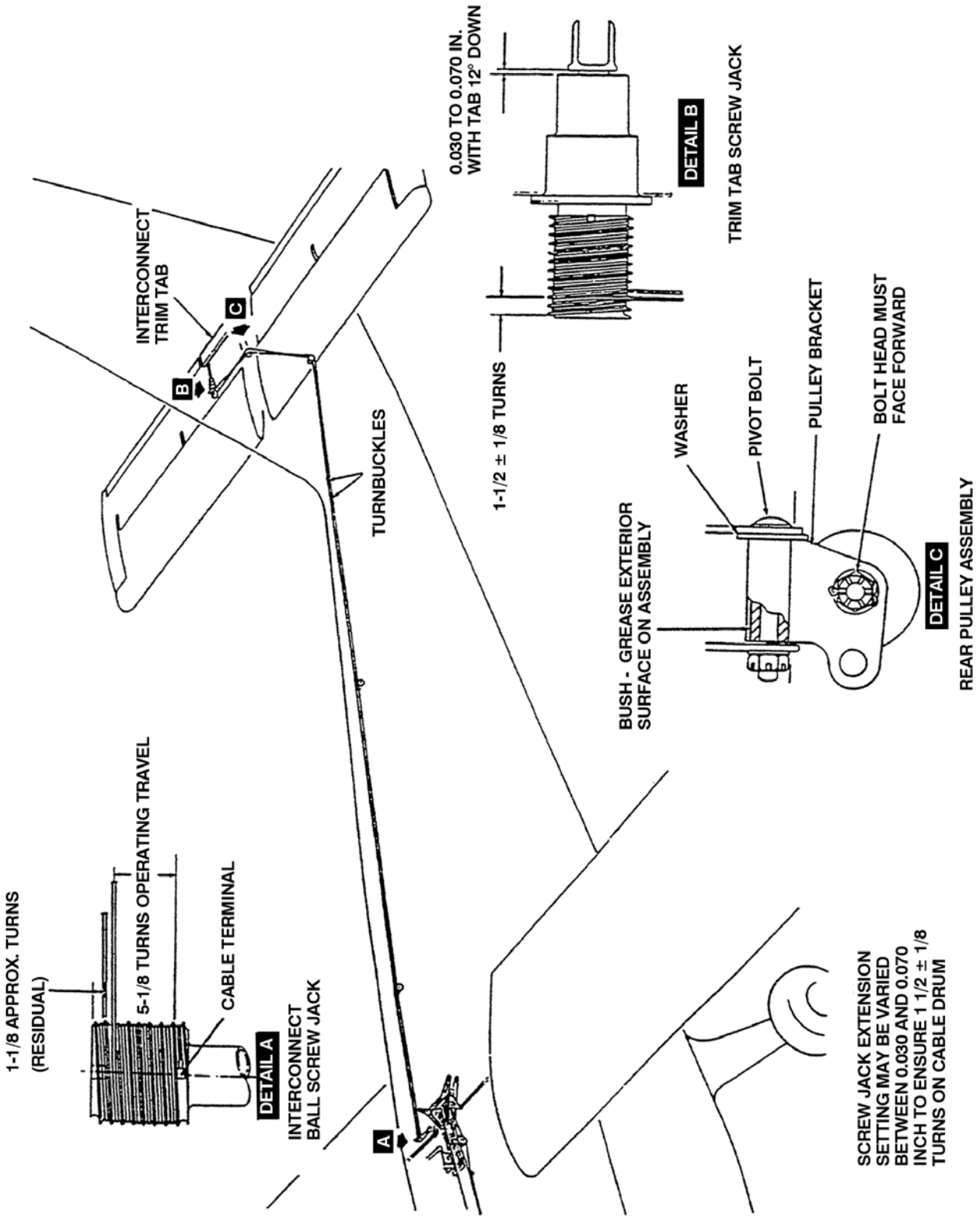
6) Set the trim tab 12° down. Measure angular difference along upper skin, or measure 0.85 ± 0.05 inch from elevator trailing edge to trim tab trailing edge with tab down.

7) With trim screw jack set as in Para 5.A.3) and tab set as in Para 5.A.6), adjust and connect trim tab connecting rod. Lock connecting rod lock nut.

8) Pump flaps fully down, ensure that at extremities of travel, ball screw jack and trim screw jack cable drums have a minimum of half a turn of cable left on drum.

9) Check flap/elevator interconnect trim tab control surface clearances (refer to 27-00-00, Flight Controls – General Data).

10) Operate flaps up and down and check interconnect system for full and free range of movement.



Flap/Elevator Trim Interconnect System – Rigging (Mod 6/1219)
Figure 1

Print Date: 2025-09-12

B. Rig Flap/Elevator Interconnect Trim System (Mod 6/1775)

Note Before attempting to rig the flap/elevator interconnect trim system, ensure that the wing flap system is correctly rigged. (Refer to 27-50-00, Wing Flaps System – General Data).

Refer to Figure 2 for details.

1) Ensure wing flaps are pumped fully up.

2) Set cables on interconnect ball screw jack cable drum as in Figure 2, "Detail A", with inboard cable terminal uppermost.

3) Disconnect connecting rod between screw jack and tab attachment bracket at tab attachment.

4) Set trim screw jack extension to 0.050 inch and arrange cables on screw jack drum as shown in Figure 2, "Detail B".

5) Connect turnbuckles and tighten to approximately equal lengths.

6) Check there is clearance at trim tab screw jack as shown in Figure 2, "Detail B", of 0.030/0.070 inch. If faces touch, dismount jack, rotate its mounting flange 180° and reinstall jack.

7) Measure clearance at trim tab screw jack as shown in Figure 2, "Detail B".

8) If clearance is between 0.030 and 0.070 inch, omit Para 5.B.9) or Para 5.B.10).

9) If clearance is between 0.071 and 0.084 inch:

   a) Dismount ball screw jack (Figure 2, "Detail A"), and rotate 180° clockwise (looking outboard). Transfer the 0.100 inch thick packing from outboard to inboard side of structure, to place it under jack.

   b) Dismount trim tab screw jack (Figure 2, "Detail B"), and rotate 180° counterclockwise (looking aft).

   c) Reinstall both jacks.

10) If clearance is between 0.001 and 0.029 inch:

   a) Dismount ball screw jack (Figure 2, "Detail A"), and rotate 180° clockwise (looking outboard). Reinstall 0.100 inch thick packing on outboard side of structure.

   b) Dismount trim tab screw jack (Figure 2, "Detail B"), and rotate 180° counterclockwise (looking aft).

   c) Reinstall both jacks.

11) Adjust turnbuckles to obtain correct cable tension values. For cable tension values, refer to 27-00-00, Flight Controls – General Data. Maintain trim jack setting obtained in Para 5.B.9) or Para 5.B.10).

12) Lock both turnbuckles.

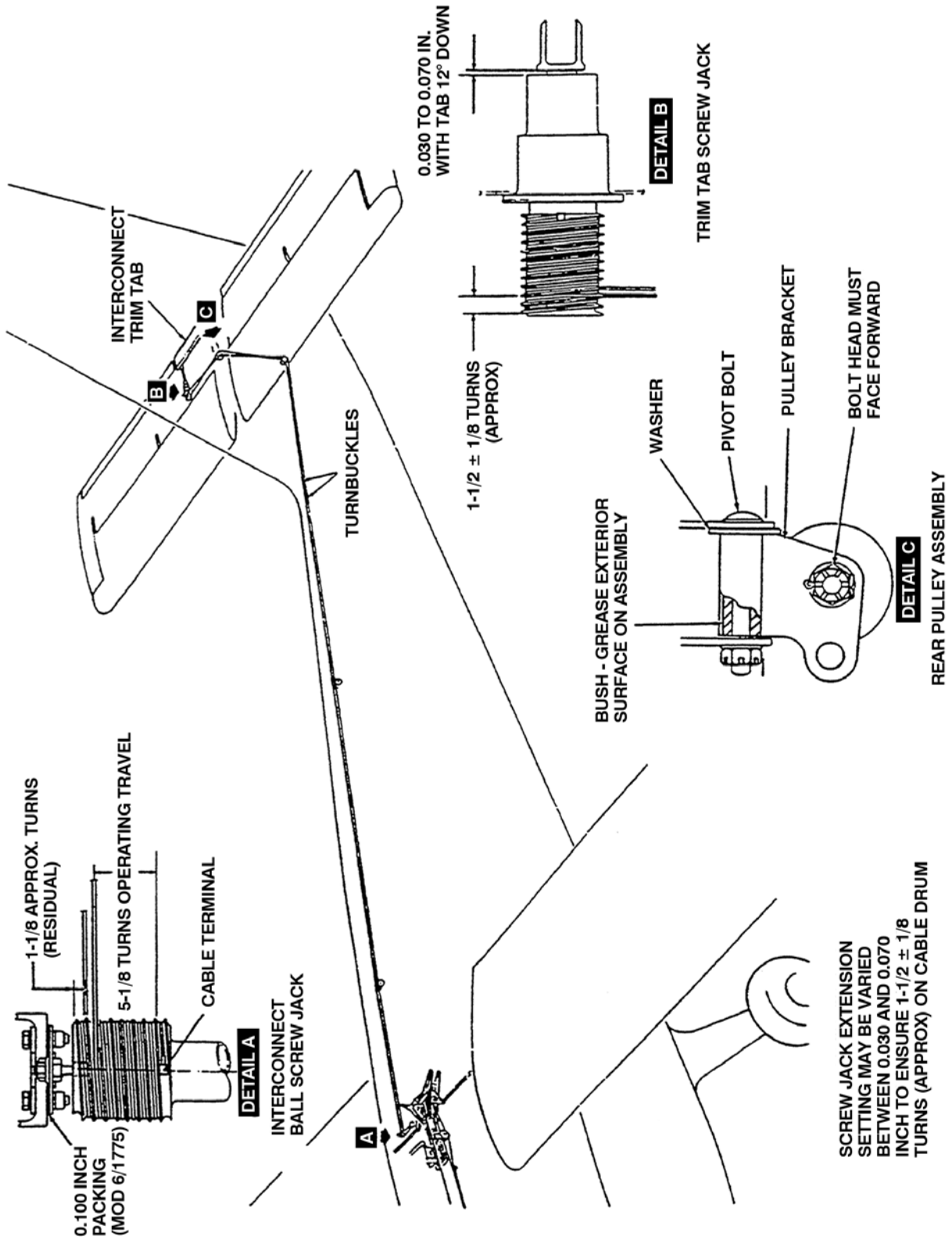
13) Set the trim tab 12° down. Measure angular difference along upper skin, or measure 0.85 ± 0.05 inch from elevator trailing edge to trim tab trailing edge with tab down.

14) With trim screw jack set as in Para 5.B.9) or Para 5.B.10) and tab set as in Para 5.B.13), adjust and connect trim tab connecting rod. Lock connecting rod lock nut.

15) Pump flaps fully down, ensure that at extremities of travel, ball screw jack and trim screw jack cable drums have a minimum of half a turn of cable left on drum.

16) Check flap/elevator interconnect trim tab control surface clearance (refer to 27-00-00, Flight Controls – General Data).

17) Operate flaps up and down and check interconnect system for full and free range of movement.



Flap/Elevator Trim Interconnect System – Rigging (Mod 6/1775)
Figure 2

Print Date: 2025-09-12

6. Inspection/Check

A. Free Play Check – Flap/Elevator Interconnect Tab

1) Select and pump flaps slowly down until flap/elevator interconnect tab is aligned with elevator trailing edge.

2) Operate elevators to neutral position.

3) Move trim tab manually and check that free play (backlash), measured at trailing edge, does not exceed 0.10–inch.

Note Measurement to be taken adjacent to control rod attachment point.

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